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1 Stochastic Bilevel Model

1.1 Lower-Level Problem (LLP)

The lower-level problem minimizes the total expected cost of covering demand, using supply from the microgrid owner, supply from the public grid, deterministic demand response, and costly critical demand suppression.

$$\min_{\substack{q_t^{cov,MGO}, q_{t,s}^{cov,PG}, \\ q_t^{supp}, q_{t,s}^{supp,cri}, \Delta d_t}} \quad C^{MGO} + C^{PG} + C^{VOLL} \tag{1}$$

The MGO cost component is the payment for the non-scenario-dependent quantity accepted from the microgrid owner at the offered price.

$$C^{MGO} = \sum_{t \in T} \tilde{p}_t \, q_t^{cov,MGO} \tag{2}$$

The public grid cost component is the expected payment for the quantity supplied by the public grid.

$$C^{PG} = \sum_{s \in S} \sum_{t \in T} prob_s \, p_t^{PG} \, q_{t,s}^{cov,PG} \tag{3}$$

The critical suppression cost component is the expected penalty for unmet critical demand valued at the value of lost load.

$$C^{VOLL} = \sum_{s \in S} \sum_{t \in T} prob_s \, p^{VOLL} \, q_{t,s}^{supp,cri} \tag{4}$$

The demand balance requires shifted demand to be covered by MGO supply, public grid supply, deterministic non-critical demand suppression, and critical demand suppression.

$$D_t + \Delta d_t = q_t^{cov,MGO} + q_{t,s}^{cov,PG} + q_t^{supp} + q_{t,s}^{supp,cri} \qquad \forall t \in T,\ s \in S \tag{5}$$

The deterministic demand-response constraint fixes non-critical suppressed demand as a function of the relative offered price and follows the demand-response behavior from the original Lebanese microgrid paper [1].

$$q_t^{supp} = \frac{\tilde{p}_t}{P_t} \, \alpha_1 \, D_t \qquad \forall t \in T \tag{6}$$

The public grid supply is limited by the scenario-dependent available public grid capacity.

$$q_{t,s}^{cov,PG} \leq \overline{Q}_{t,s}^{PG} \qquad \forall t \in T,\ s \in S \tag{7}$$

The shiftable demand in each time period is bounded by ten percent of the original demand.

$$\Delta d_t \leq 0.1 D_t \qquad \forall t \in T \tag{8}$$

The negative shiftable demand in each time period is bounded by ten percent of the original demand.

$$\Delta d_t \geq -0.1 D_t \qquad \forall t \in T \tag{9}$$

The net demand shift over the full time horizon is zero.

$$\sum_{t \in T} \Delta d_t = 0 \tag{10}$$

All lower-level supply and suppression quantities are non-negative.

$$q_t^{cov,MGO} \geq 0 \qquad \forall t \in T \tag{11}$$

$$q_{t,s}^{cov,PG} \geq 0 \qquad \forall t \in T,\ s \in S \tag{12}$$

$$q_t^{supp} \geq 0 \qquad \forall t \in T \tag{13}$$

$$q_{t,s}^{supp,cri} \geq 0 \qquad \forall t \in T,\ s \in S. \tag{14}$$

1.2 Upper-Level Problem (ULP)

The upper-level problem maximizes the microgrid owner's profit as the MGO revenue minus short-term generation costs.

$$\max_{\tilde{p}_t,\, q_t^{diesel},\, q_t^{PV}} \quad \Pi^{MGO} = \sum_{t \in T} \tilde{p}_t\, q_t^{cov,MGO} - C^{diesel} - C^{PV} \tag{15}$$

The MGO price offer is bounded between zero and an exogenous price cap.

$$0 \leq \tilde{p}_t \leq \widehat{P}_t \qquad \forall t \in T \tag{16}$$

The diesel generation cost component captures the dispatch cost of the diesel generator.

$$C^{diesel} = \sum_{t \in T} c_t^{diesel}\, q_t^{diesel} \tag{17}$$

The PV generation cost component captures the dispatch cost of PV generation, although this cost is probably zero and may therefore be negligible.

$$C^{PV} = \sum_{t \in T} c_t^{PV}\, q_t^{PV} \tag{18}$$

The accepted MGO quantity must be supplied by diesel generation and PV generation.

$$q_t^{cov,MGO} = q_t^{diesel} + q_t^{PV} \qquad \forall t \in T \tag{19}$$

The diesel generator is dispatchable up to its installed capacity.

$$q_t^{diesel} \leq \widehat{Q}^{diesel} \qquad \forall t \in T \tag{20}$$

The PV generation is limited by the time-dependent available PV generation cap.

$$q_t^{PV} \leq \widehat{Q}_t^{PV} \qquad \forall t \in T \tag{21}$$

The upper-level generation quantities are non-negative.

$$q_t^{diesel} \geq 0 \qquad \forall t \in T \tag{22}$$

$$q_t^{PV} \geq 0 \qquad \forall t \in T. \tag{23}$$

2 KKT Reformulation of the Follower

References

[1] S. Zwickl-Bernhard, A. F. Neumann, M. Olleik, and H. M. Dbouk, "Market organization in low-income countries' microgrids: Insights from electricity demand elasticity and game-theory optimization. case study: Lebanon," *IEEE Transactions on Energy Markets, Policy and Regulation*, pp. 1–10, 2025. DOI: 10.1109/TEMPR.2025.3585959. [Online]. Available: https://doi.org/10.1109/TEMPR.2025.3585959 (cit. on p. 2).